

Economics 2220

Principles Of Macroeconomics

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Lecture 10:

The Aggregate Expenditure Model

► What you will learn in this chapter



- How the inventory adjustment process moves the economy to a new equilibrium after a change in demand
- Why investment spending is considered a leading indicator of the future state of the economy

THE INCOME-EXPENDITURE MODEL



A more detailed look at how changes in spending get multiplied and how firms respond.

Inventories play a central role in the economy, so economists pay attention to the changes in firms' inventories when trying to understand the direction of the economy.

THE INCOME–EXPENDITURE MODEL

The assumptions we make:

Changes in overall spending lead to changes in aggregate output.

The interest rate is fixed.

Taxes, government transfers, and government purchases are all zero.

Exports and imports are both zero.

(Future chapters loosen these assumptions.)

PLANNED AGGREGATE SPENDING AND REAL GDP

In our simplified economy,

$$\mathbf{GDP = C + I \text{ and}}$$

$$\mathbf{YD = GDP}$$

and our aggregate consumption function is

$$\mathbf{C = A + MPC \times YD}$$

and we assume I_{planned} is fixed

so

$$\mathbf{AE_{Planned} = C + I_{Planned}}$$

where **planned aggregate spending** = the total amount of planned spending in the economy.

WORKING WITH THE MODEL

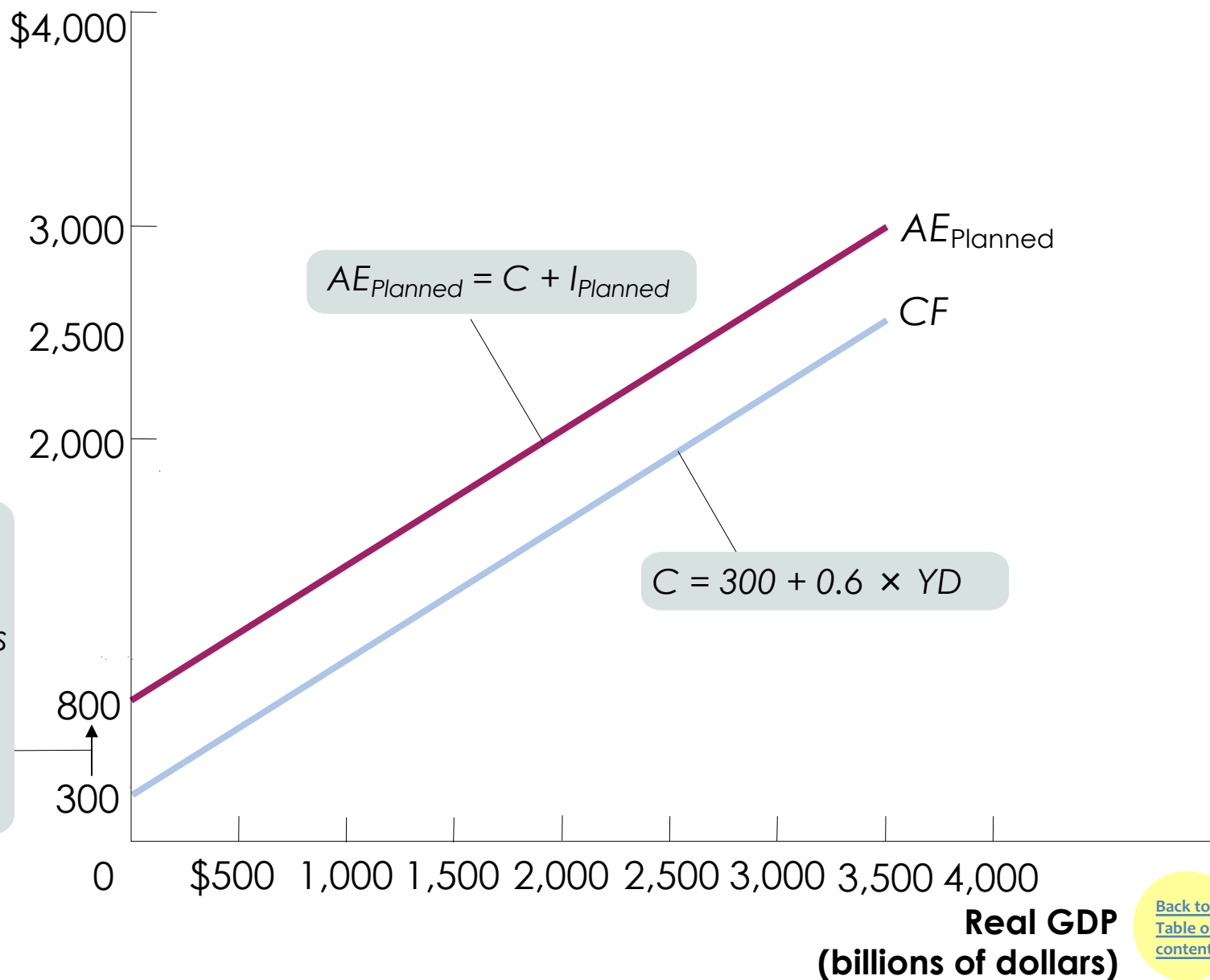
Example of the model in action:

Example: if $C = 300 + 0.6 \times YD$, then

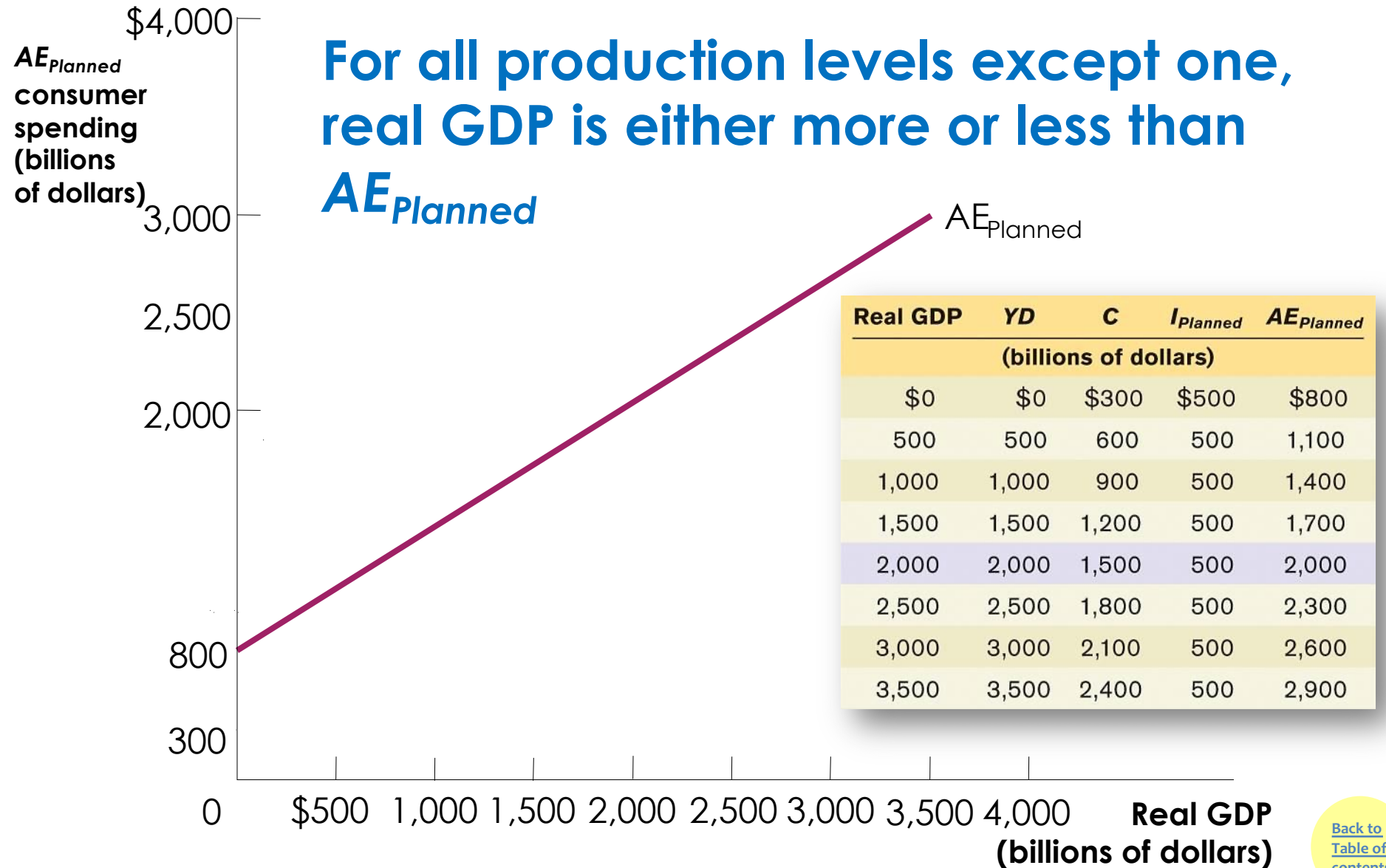
Real GDP	YD	C	$I_{Planned}$	$AE_{Planned}$
(billions of dollars)				
\$0	\$0	\$300	\$500	\$800
500	500	600	500	1,100
1,000	1,000	900	500	1,400
1,500	1,500	1,200	500	1,700
2,000	2,000	1,500	500	2,000
2,500	2,500	1,800	500	2,300
3,000	3,000	2,100	500	2,600
3,500	3,500	2,400	500	2,900

PLANNED AGGREGATE SPENDING AND GDP

$AE_{Planned}$
consumer
spending
(billions
of dollars)



PLANNED AGGREGATE SPENDING AND GDP



INCOME–EXPENDITURE EQUILIBRIUM

Planned aggregate spending can be different from real GDP only if there is unplanned inventory investment, $I_{Unplanned}$, in the economy.

*If firms have **overestimated sales** and produced too much, there will be unintended additions to inventories (**and $I_{Unplanned}$ will be positive**).*

*If firms have **underestimated sales** and produced too little, there will be unintended drops in inventories (**and $I_{Unplanned}$ will be negative**).*

LEARN BY DOING: PRACTICE QUESTION

Suppose the level of planned aggregate expenditure in an economy is \$500 and real GDP is \$600. According to the our model:

- a) unplanned increases in inventories are occurring.**
- b) unplanned decreases in inventories are occurring.**
- c) inventories will be unaffected and will remain at the planned inventory level.**
- d) excess production will continue, since there is no mechanism to restore the level of production to the level of spending.**

INCOME-EXPENDITURE EQUILIBRIUM

$$GDP = C + I$$

$$= C + I_{Planned} + I_{Unplanned}$$

$$= AE_{Planned} + I_{unplanned}$$

Whenever real GDP exceeds $AE_{Planned}$, $I_{Unplanned}$ is positive.

Whenever real GDP is less than $AE_{Planned}$, $I_{Unplanned}$ is negative.

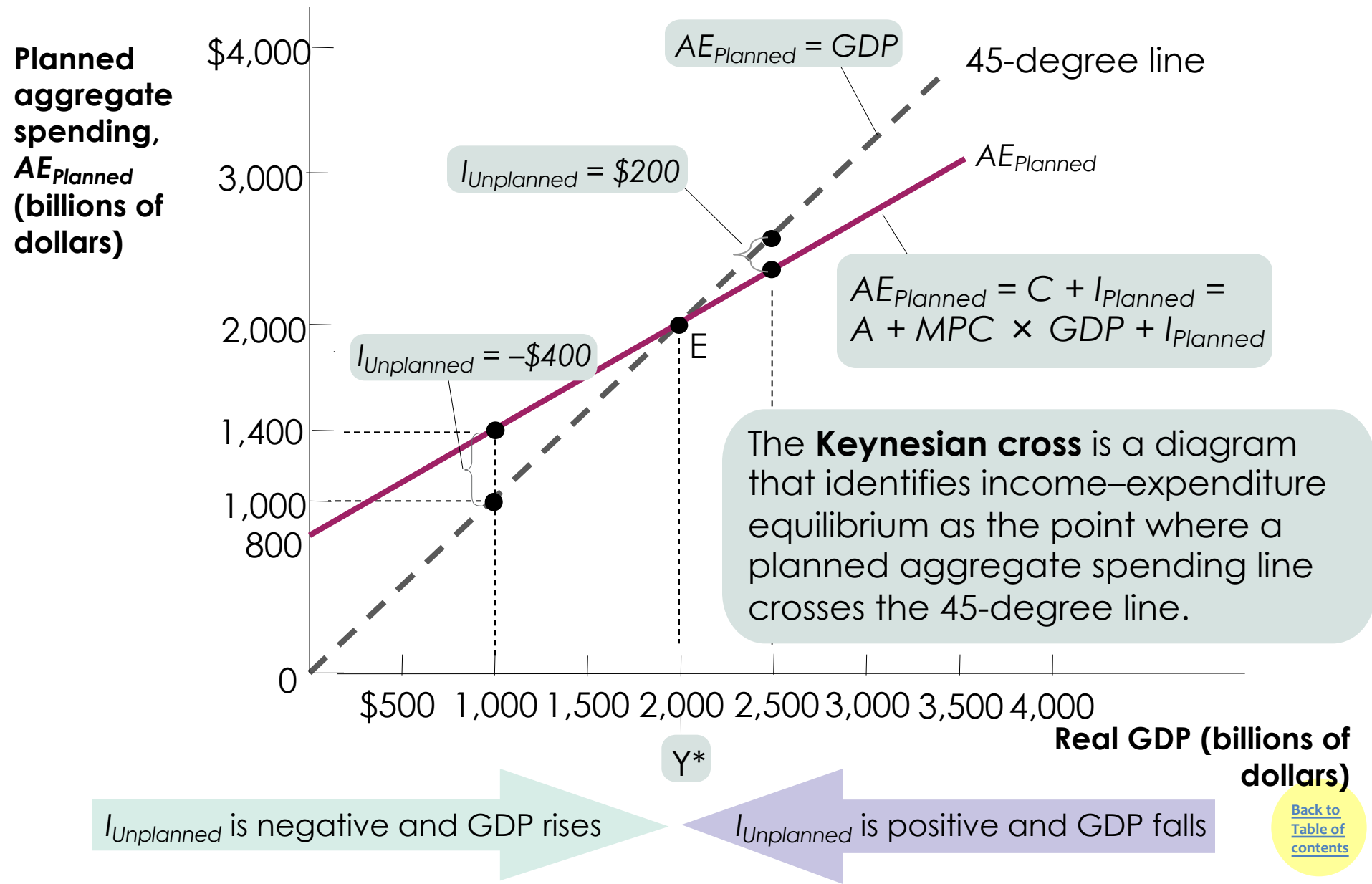
(But firms will act to correct their mistakes by producing more or less accordingly.)

INCOME-EXPENDITURE EQUILIBRIUM

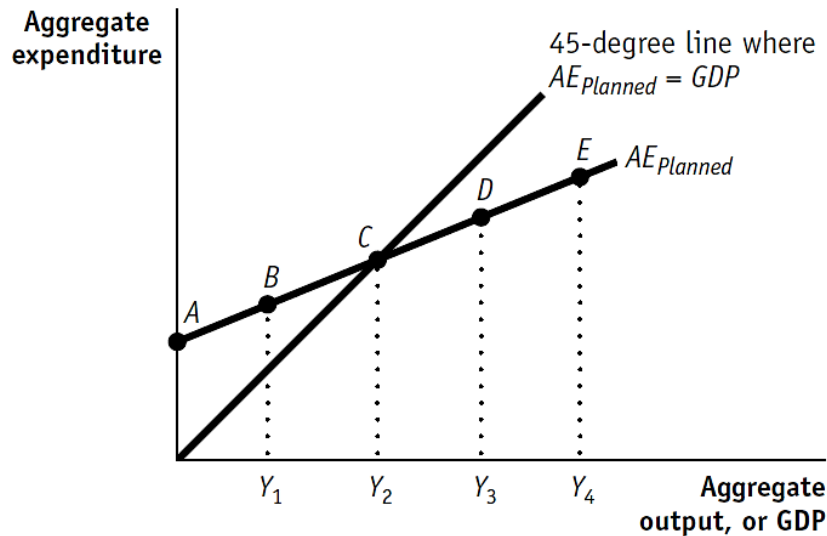
The economy is in **income-expenditure equilibrium** when aggregate output (real GDP) is equal to planned aggregate spending.

Income-expenditure equilibrium GDP: the level of real GDP at which real GDP equals planned aggregate spending.

INCOME-EXPENDITURE EQUILIBRIUM



LEARN BY DOING: DISCUSS



Suppose Outlandia's economy is operating at point B.

- Is this an income –expenditure equilibrium for Outlandia? Describe what is happening to inventories and production in Outlandia.
- What level of GDP will this economy produce at equilibrium?
- At what levels of GDP is unplanned inventory investment positive?
- At what levels is planned aggregate expenditure greater than income?

LEARN BY DOING: DISCUSS

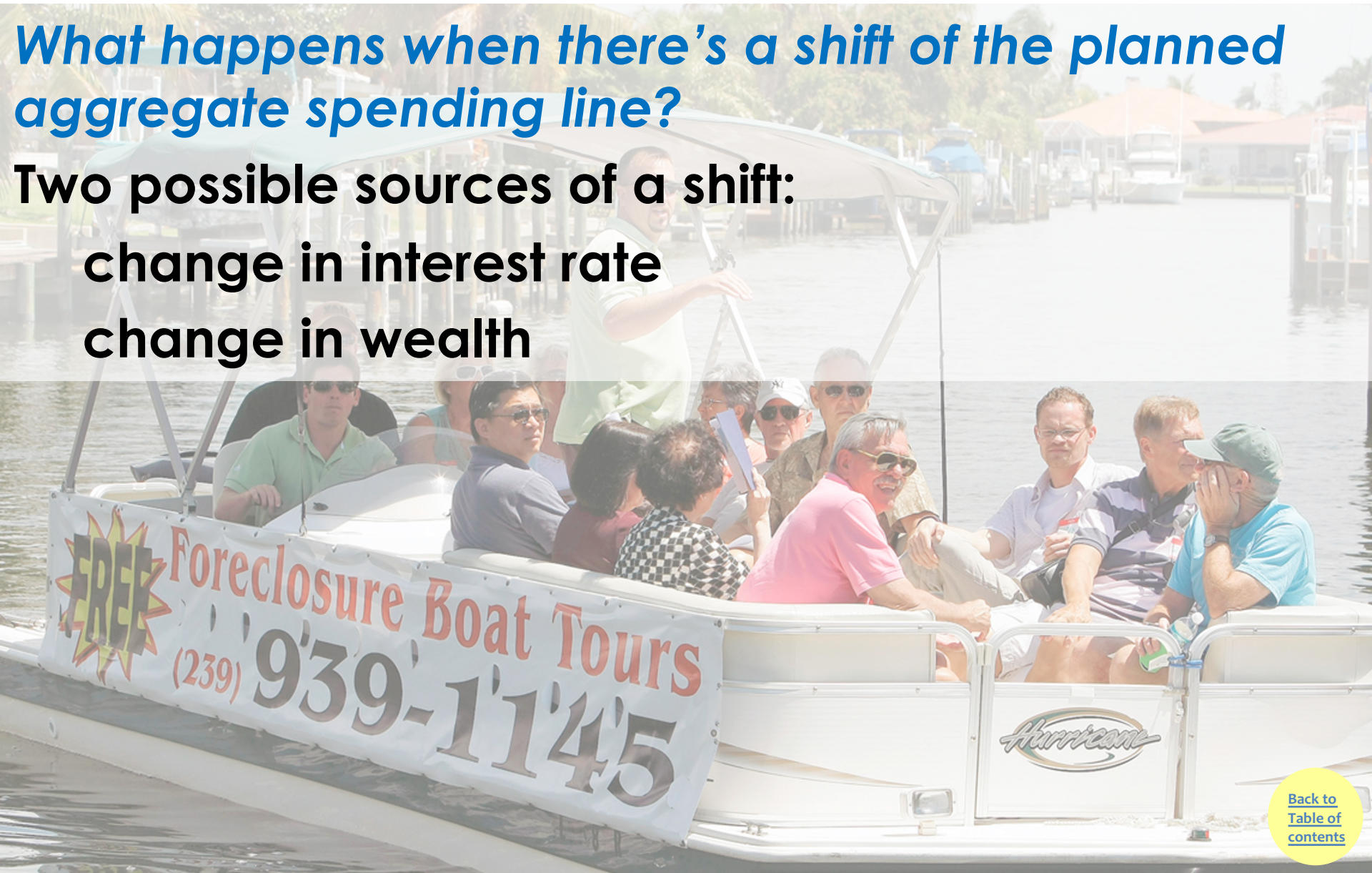
(answers to previous slide's questions)

- a) No, the income–expenditure equilibrium is at point C on the graph. At point B, $AE_{Planned}$ is greater than GDP, or income. This causes inventories to fall, and firms view this as a signal that they should increase their production from Y_1 toward Y_2 , the income–expenditure equilibrium real GDP.
- b) This economy, holding everything else constant, will produce GDP equal to Y_2 when it is in income–expenditure equilibrium.
- c) Unplanned inventory investment is positive wherever GDP is greater than $AE_{Planned}$: GDP output levels Y_3 and Y_4 represent two output levels where unplanned inventory investment is positive. So at any output level greater than Y_2 , unplanned inventory investment is positive.
- d) This is true for any level of GDP less than Y_2 .

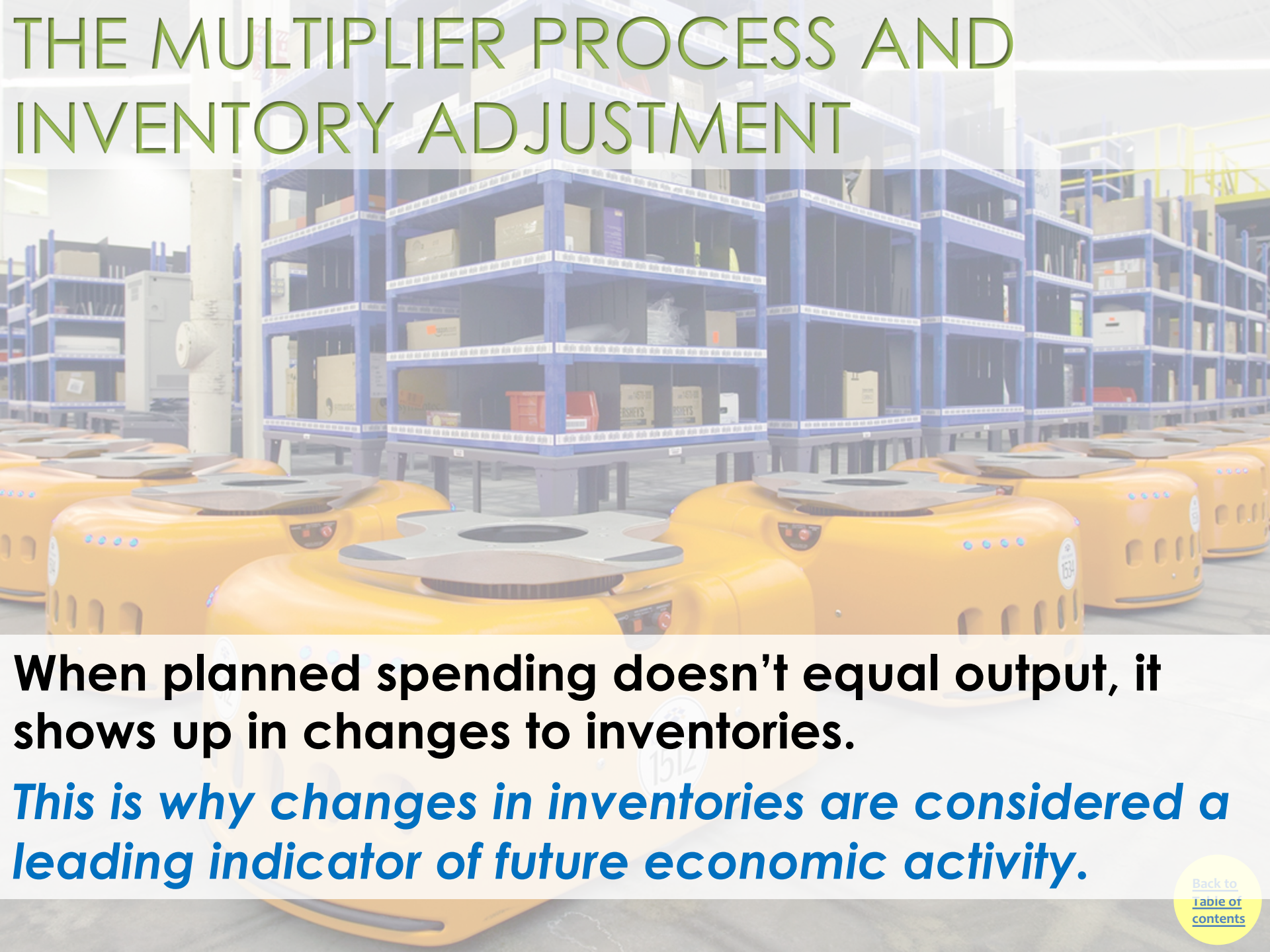
THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

What happens when there's a shift of the planned aggregate spending line?

Two possible sources of a shift:
change in interest rate
change in wealth



THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

A photograph of a warehouse interior. In the foreground, several yellow autonomous mobile robots (AMRs) are parked in rows. They have a rounded, boxy design with a grey top surface and blue lights. In the background, there are tall blue metal shelving units (pallet racks) filled with cardboard boxes. The floor is a light-colored concrete.

When planned spending doesn't equal output, it shows up in changes to inventories.

This is why changes in inventories are considered a leading indicator of future economic activity.

THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

A Vivid Example: Mario's Pizzeria

Every Friday, Mario must decide how many pizzas to bake before he knows how many customers will show up.

Production (pizzas baked) = Real GDP

Actual purchases by customers = AE_{Planned}

Unsold pizzas (or emergency freezer raids) = $I_{\text{Unplanned}}$

What happens when Mario guesses wrong?

THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

Scenario 1: Mario Overestimates Demand

Mario expects a busy Friday and bakes 200 pizzas.
A big storm hits — customers only buy 150 pizzas.
50 unsold pizzas pile up in the warming racks.

$I_{\text{Unplanned}} = +50$ (*unintended additions to inventory*)

Production (200) > Spending (150)

→ Economy is *above equilibrium*

Next Friday, Mario will **cut back production**

→ Output shrinks toward equilibrium.

THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

Scenario 2: Mario Underestimates Demand

Mario expects a slow night and bakes only 100 pizzas. A nearby concert lets out early — customers want 140 pizzas.

Mario sells all 100 and pulls 40 frozen pizzas from his back freezer.

$$I_{\text{Unplanned}} = -40 \text{ (unintended drops in inventory)}$$

Production (100) < Spending (140)

→ Economy is *below equilibrium*

Next Friday, Mario will **ramp up production**

→ Output rises toward equilibrium.

THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

Scenario 3: Mario Nails His Forecast

Bakes 150 pizzas, exactly 150 sell. $I_{\text{Unplanned}} = 0$. Economy is in equilibrium.

	Prod > Spending	Prod < Spending	Prod = Spending
Inventories	Pile up	Drain	Stay as planned
$I(\text{Unplanned})$	Positive (+)	Negative (–)	Zero
Firms respond	Cut production	Raise production	Keep steady
Economy	Down toward eq.	Up toward eq.	At equilibrium

THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

Unplanned inventory changes are the signal that tells firms they've guessed wrong.

They are the mechanism that pushes the economy back toward equilibrium:

$AE_{\text{Planned}} < \text{GDP} \rightarrow I_{\text{Unplanned}} > 0 \rightarrow \text{Firms cut production} \rightarrow \text{GDP falls}$

$AE_{\text{Planned}} > \text{GDP} \rightarrow I_{\text{Unplanned}} < 0 \rightarrow \text{Firms increase production} \rightarrow \text{GDP rises}$

$AE_{\text{Planned}} = \text{GDP} \rightarrow I_{\text{Unplanned}} = 0 \rightarrow \text{Equilibrium!}$

Just like Mario adjusting how many pizzas he bakes next Friday!

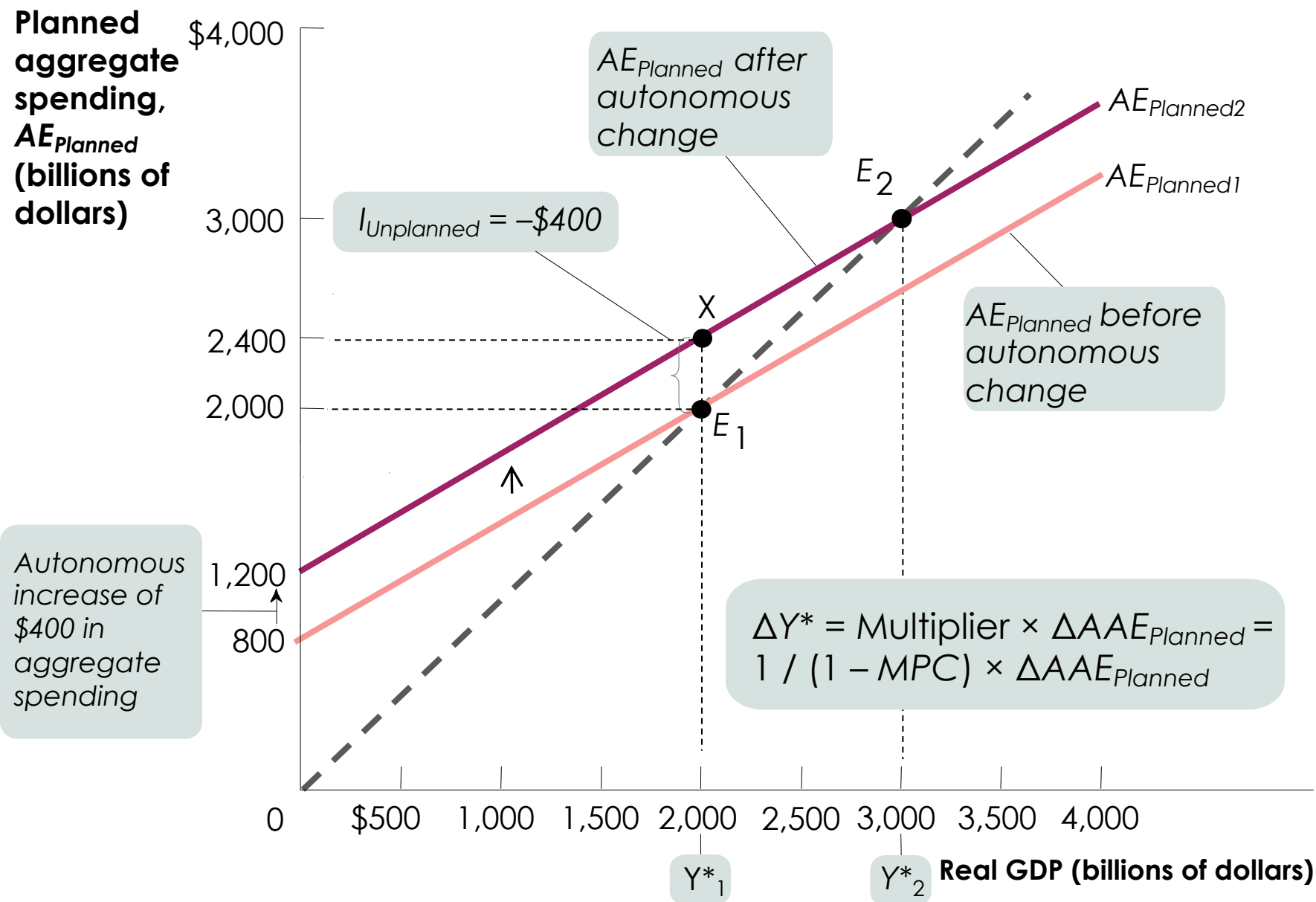
THE MULTIPLIER PROCESS AND INVENTORY ADJUSTMENT

Continuing our previous example: the effect of an autonomous increase in planned aggregate spending of \$400 billion

Real GDP	$AE_{Planned}$ before autonomous change	$AE_{Planned}$ after autonomous change
(billions of dollars)		
\$0	\$800	\$1,200
500	1,100	1,500
1,000	1,400	1,800
1,500	1,700	2,100
2,000	2,000	2,400
2,500	2,300	2,700
3,000	2,600	3,000
3,500	2,900	3,300
4,000	3,200	3,600

The fact that the rise in equilibrium GDP, from \$2,000 to \$3,000, is much larger than the autonomous increase in aggregate spending, (\$400) has a familiar explanation: the multiplier process.

THE MULTIPLIER



THE PARADOX OF THRIFT

In macroeconomics, the outcome of many individual actions can generate a result that is different from and worse than the simple sum of those individual actions.

If your house loses value because the housing bubble has burst, you might adjust and spend less.

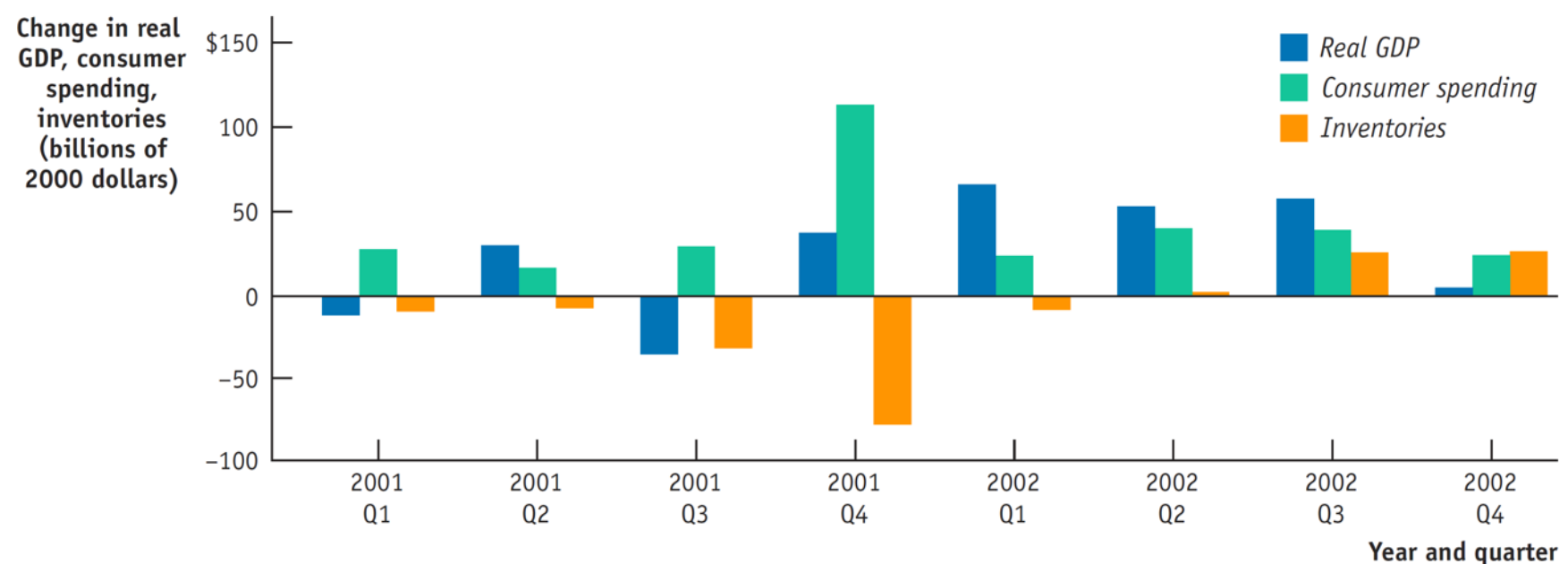
If everyone else does the same, the economy is depressed, jobs are lost, and everyone is worse off because of their virtuous individual actions.

Veiman/Marx

INVENTORIES AND THE END OF A RECESSION

A look at the multiplier process at the end of a recession

Notice the huge increase in consumer spending Q4 2001 that caught businesses by surprise—but by Q1 2002 businesses increased production and real GDP jumped.



Source: Bureau of Economic Analysis.

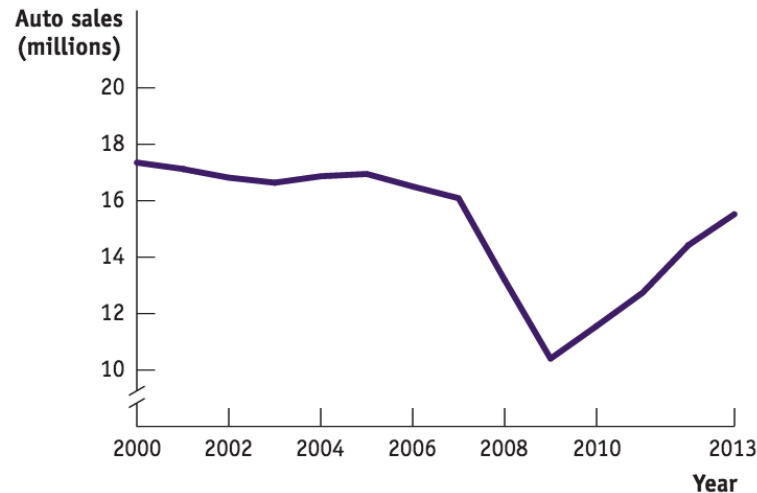
LEARN BY DOING: BUSINESS CASE

What's Good for America is Good for GM

In 2009 the U.S. economy was in a nose-dive... and so was General Motors. The government bailed out GM and took a temporary 61% ownership.

Was GM too big to fail (would letting it fall into bankruptcy cost many more jobs?)

Were GM's problems self-made or was it a victim of a poor economy?



Source: Federal Reserve Bank of St. Louis.